Toward Earlier Discovery of Urban Flood Risk in Bangkok

A spatial triage prototype, retrospective evidence, and a path to BMA field validation

Bangkok flood modeling project | Hackathon working paper | 25 October 2026

Abstract

Urban flood preparedness requires choosing where to inspect before storms, yet city teams cannot survey every street. Past complaints identify recurring trouble spots but offer less guidance for places with no recent report. We present a daily 250 m priority map for Bangkok that combines report history with terrain, roads, canals, and land cover. A site-trained convolutional neural network learns a correction to a strong history ranking. We tested the rankings against first-in-year flood-report sites recorded in Traffy during Q4 2025 across districts held out from fitting and model selection. Among 393 new sites in squares with no prior-year report, screening the top 5% captured 14.0% of sites versus 6.8% for history alone, a twofold gain of about 28 emerging sites at equal area coverage. Earlier discovery in dense commercial or transport areas could limit repair, business, and travel losses if it enables effective action. Illustrative scenarios show the economic value at stake.

Keywords urban flooding; spatial ranking; emerging sites; field triage; Bangkok

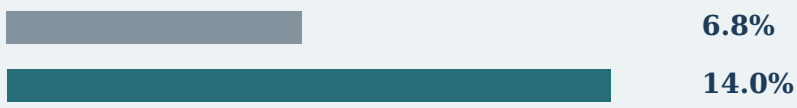
1 Introduction

Bangkok's FY2026 (October 2025–September 2026) drainage budget and annual action plan set substantial spending and targets for pipes, water control, canals, and known flood-risk points [2,3]. In its first quarter (calendar Q4 2025), our grouping of Traffy reports finds 2,326 flood-site events across Bangkok, including 471 first-in-year sites [1,8]. The proposed spatial model could help districts prioritize checks beyond familiar hotspots before the next storm.

The ranking question is concrete: as a district reviews more map squares, which new trouble spots enter the list first? Past reports are a useful starting point [4]. We test whether the proposed model ranks newly reported locations higher at equal numbers of unfamiliar squares, then identify what BMA would need to check before using it in field decisions.

2× NEW-SITE CAPTURE AT THE SAME 5% UNFAMILIAR-AREA COVERAGE

Both methods rank the same 250 m squares with no report in the prior year.
Of 393 later new report sites, share shortlisted:



Proposed model: history score + CNN using report maps, terrain, roads, canals, and cover.
Full district top 5% of all cells: both find 1.27% of 471 new sites.

2 Study area, data, and target

Bangkok's 250 m cells combine dated Traffy reports, SRTM terrain, OpenStreetMap roads and canals, and ESA WorldCover 2021 [5-7]. These predict reported sites, not observed water. The convolutional neural network (CNN) has no rain input.

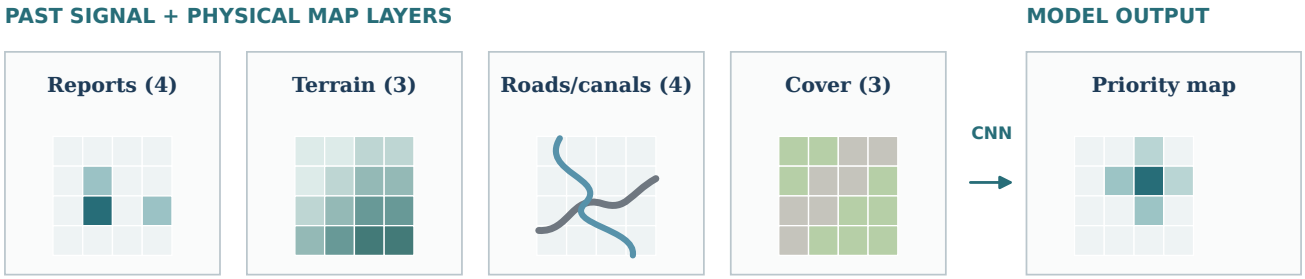


Figure 1. Schematic 250 m grids: four report, three terrain, four road/canal, three cover, and two mask channels form 16 CNN inputs. Colors illustrate ranking, not observed water.

2.1 Defining a first-in-year reported site

Most of the 2,326 Q4 site events revisit familiar places. A model trained to find unfamiliar places needs a separate test on new sites. We therefore define first-in-year report-site events, then compare rankings among cells with no own-cell report history.

Nearby same-day report points form complete-linkage groups; a medoid selects one evaluation cell. A group is 'new' if none of its points lies within 250 m of an earlier report in the preceding 365 days [8]. Counting each group once avoids complaint-buffer spillover into adjacent cells. Q4 has 471 new site events; 393 have no own-cell history.

2.2 Ranking evaluation and outcome

Before rain, crews cannot check every area; the proposed value is deciding which unfamiliar places to examine first. We rank all cells with no own-cell report in the previous 365 days within each district and day. Of 471 Q4 first-in-year site events, 393 fall in this candidate set. We ask how highly each later site medoid ranks, and show capture as the review fraction grows. No share of staff time is assigned to discovery by this test [11].

The percentile rank is the share of unfamiliar candidates scoring below a reported site; higher is better. The capture curve shows the share of those 393 site events in the top fraction of unfamiliar candidates. Section 5.3 checks a full district queue without reserved slots. A practical field test would set stops from visit logs and measure verified findings per crew-hour. Training uses earlier site medoids and unlabeled same-day rivals (Section 4); prevented floods are not measured.

HOW THE RETROSPECTIVE TEST IS SEPARATED

	2023-2024	Jan-Sep 2025	Oct-Dec 2025
40 source districts	Train model	Choose settings	--
10 held-out districts	No fitting	No selection	Test new sites

Figure 2. Five district folds; held-out outcomes do not fit or select the model.

3 Spatial ranking model

For candidate cell i and day t, a 15 x 15 patch X(i,t) contains 16 aligned spatial channels. Each patch pixel represents a 250 m cell. History maps are built strictly from dates before t; physical channels are static snapshots. Nonlinear spatial mixing permits conditional effects such as neighboring report density interacting with low terrain, canal proximity, and built cover. It does not by itself identify a hydraulic flow path.

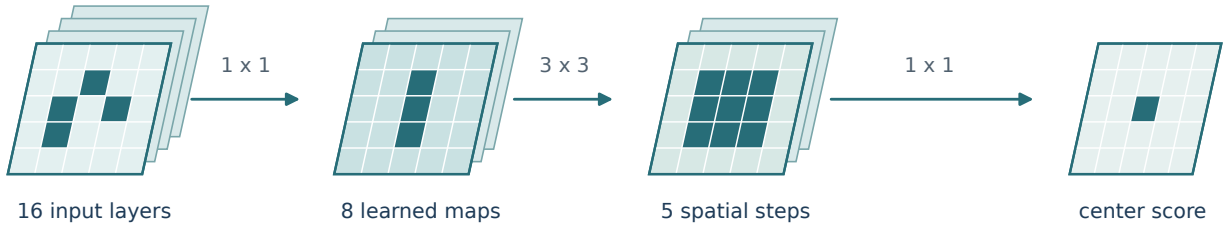


Figure 3. Perspective view of the CNN. Stacks show channels, not physical height; grids are simplified from the 15 x 15 input. The highlighted center is the score correction.

3.1 How the CNN reads a neighborhood

The 15 x 15 input patch covers about 3.75 km across. The entry layer mixes the 16 values at each cell into eight learned maps. Five spatial steps scan neighboring cells and mix those maps; the last two use wider spacing. The center output corrects the nearby-history score. This learns associations with later reports, not a simulated water route.

3.2 History reference and residual score

The unfamiliar-cell reference score is a weighted combination of log-transformed prior-year reports smoothed at approximately 250, 750, and 1,500 m: weights 0.6, 0.3, and 0.1, respectively [9]. The site CNN starts with a zero-initialized output head and learns a correction to that score. The blend and training arm are chosen on source-district mean site rank, without an inspection cutoff.

$$s_{i,t} = (1-\alpha)\,\mathrm{rank}(h_{i,t}) + \alpha\,\mathrm{rank}(h_{i,t} + f_\theta(X_{i,t}))$$

The small residual design sets a demanding test: the CNN must contribute spatial information beyond a sensible hand-built prior while preserving the reference where it is useful. Static features are normalized using source districts only. A valid-patch mask keeps scores consistent at the grid boundary.

3.3 Why this is a ranking problem

The data lack verified dry observations. Training a calibrated flood/no-flood classifier would therefore mislabel unknown cells. Instead, the model learns whether a known positive reported site should outrank plausible unreported alternatives within the same district and day. That objective matches the ranking evaluation, but its score has no direct probabilistic interpretation.

4 Training design and spatial masking

The earlier cell-trained CNN used report-positive cell-days and randomly suppressed 10% of neighboring history pixels. A later synthetic-center variant added repeat sites whose center history was hidden, sometimes together with a random 3 x 3 block. Source-selected cell-day gains were small and uncertain [9,10]. The present experiment changes the positive target to raw-point first-in-year site medoids. This target redesign and the training comparison both matter; the gain cannot be assigned to masking alone.

4.1 Two site-trained arms

The new-site-only arm trains on genuine 2023-2024 first-in-year medoids. Those positive cells naturally have zero own-cell history, so no synthetic center mask is applied. The group-mask arm adds repeat-site medoids and zeros their 3 x 3 neighborhood in all four report-history planes. Physical and geometry channels remain visible, and these synthetic positives receive 0.3 loss weight. No artificial mask is used at inference [8].

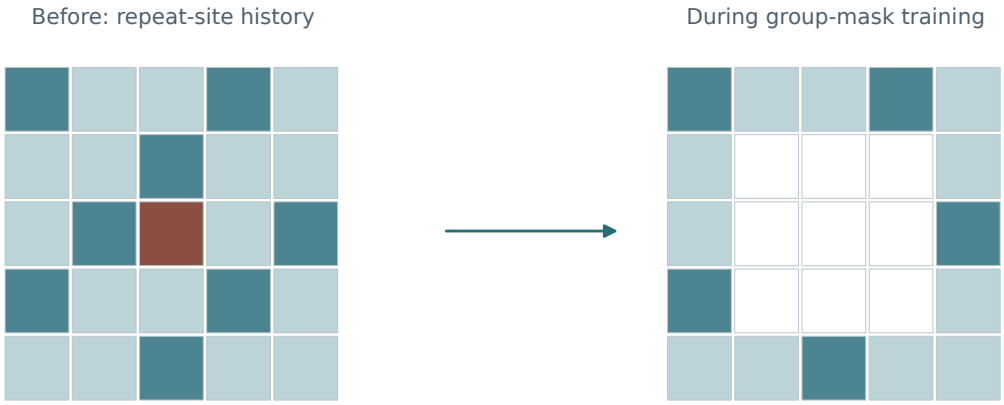


Figure 4. Schematic 5 x 5 crop from each 15 x 15 report-history patch. The optional arm hides the centered 3 x 3 history block at repeat sites. Static terrain, road, canal, and cover channels are unchanged.

4.2 Pairwise loss and comparison sampling

For each positive medoid, comparison cells are unreported, have zero own-cell history, and are in the same district on the same day. One is drawn from a similar nearby-history score; another is a high-history rival. The loss increases when a rival outranks a positive. A small squared-residual term discourages unnecessary departure from the reference score.

$$\mathcal{L} = \frac{\sum_{(i,j)} w_i \,\mathrm{softplus}(s_j - s_i)}{\sum_{(i,j)} w_i} + 0.0005\,\overline{f_\theta(X)^2}$$

Here i is a positive site, j an unlabeled rival, and w_i is 1 for genuine new-site examples or 0.3 for synthetically masked repeat sites. This is positive-unlabeled ranking, not wet/dry classification. The model can still exploit the spatial imprint of earlier reporting; masking cannot prove that it has learned water movement.

5 Geographic evaluation of emerging sites

Five geographic folds each exclude 10 Bangkok districts from weight fitting and model-arm selection. Training uses 2023-2024 source-district events. January-September 2025 source districts select the history/CNN blend and training arm by mean new-site rank percentile, without a shortlist cutoff. October-December 2025 reports in excluded districts form the Q4 evaluation [11]. Earlier test-district reports remain available as pre-day context. Source selection chooses the new-site-only arm in all five folds.

5.1 Ranking unfamiliar candidates

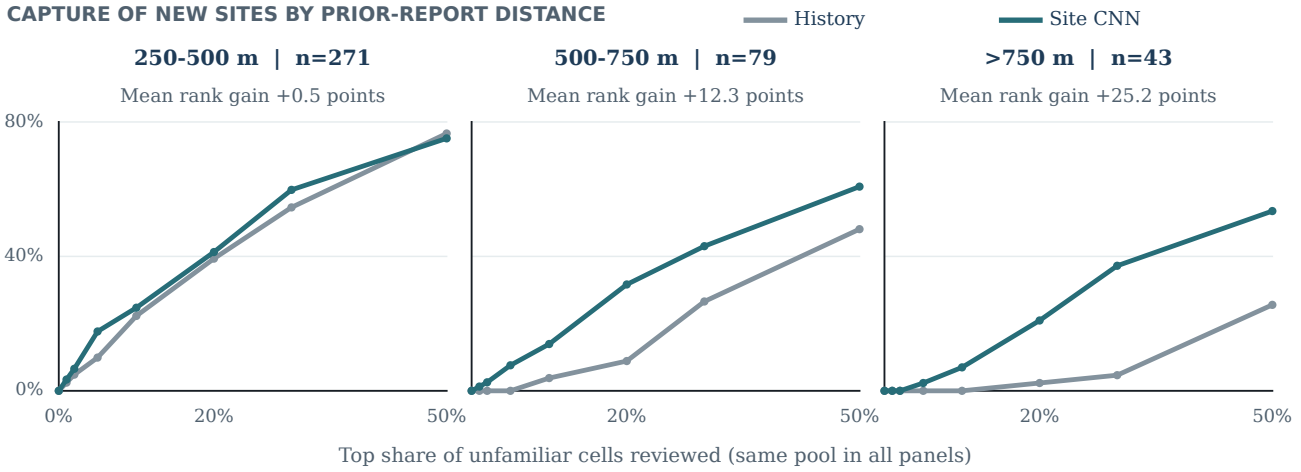


Figure 5. Q4 first-in-year reported-site capture by nearest earlier report distance. The three bands partition 393 sites; every panel ranks the same zero-own-history cells.

Across all 393 eligible new sites, mean rank percentile rises from 60.17 for nearby history to 65.70 for the source-selected CNN/history blend: +5.53 points. A paired district bootstrap interval is +2.75 to +8.00 points [11]. At 5% of unfamiliar candidates reviewed, capture is 6.82% versus 13.96%; at 10%, it is 16.13% versus 20.61%. These cutoffs illustrate the curve; none is a measured crew budget.

5.2 The gain grows with distance from past reports

History ranks new sites well near older reports. The CNN's mean-rank gain rises with distance: +0.5 points for 271 sites 250-500 m away, +12.3 for 79 sites 500-750 m away, and +25.2 for 43 sites beyond 750 m [16]. At a 30% unfamiliar-cell screen, capture beyond 750 m is 4.7% for history versus 37.2% for the CNN. Figure 5 shows the distance-stratified curves. The pattern suggests value where report history is sparse; it does not prove water-flow learning or define a crew budget.

5.3 The full-district shortlist

The 5% test above compares rankings within unfamiliar cells. In a full district list where all cells compete, both rankers capture 24.71% of all 2,326 reported sites and 1.27% of 471 emerging sites at the top 5% of all cells [15]. At the top 10%, both also tie: 38.32% of all sites and 3.50% of emerging sites. The conditional gain within unfamiliar cells has not improved this full queue. Q4 was reused in development, and reports are not verified flood extent.

6 Economic loss from an overlooked flooded square

A 250 m square covers 62,500 square metres, but its flooded footprint may be much smaller. One real flood can still impose repair, interruption, travel, and public-response costs beyond the visible water. The cost depends on depth, duration, exposed people and assets, and alternatives for travel. A Traffy report does not measure those quantities.

6.1 The same flooded area can expose very different value

A quiet side street and a dense commercial square occupy the same 250 m grid area, yet the latter can contain many premises and a busy through-route. Table 1 gives two transparent stress cases. Neither is an estimated Bangkok average or a forecast for any Q4 reported site.

Table 1. Hypothetical loss per square; prem=premises, shop-h=shop-hours, p-h=person-hours.

Loss pathway	Ordinary square	Dense commercial / arterial
Cleaning / repair	40 prem x THB5k/prem = THB0.20m	150 prem x THB40k/prem = THB6.00m
Business interruption	20 shop-h x THB2k/shop-h = THB0.04m	500 shop-h x THB8k/shop-h = THB4.00m
Through-traveler delay	400 p-h x THB100/p-h = THB0.04m	10,000 p-h x THB150/p-h = THB1.50m
Extra public response	THB0.02m	THB0.50m
Illustrative total	**THB 0.30m**	**THB 12.00m**

The dense case assumes 150 affected premises, 100 businesses losing five hours of value added, and 10,000 additional hours lost by through-travelers. All unit values are assumptions; business and traveler hours must not overlap.

6.2 Which sites enter the unfamiliar-cell shortlist?

At the top 5% of unfamiliar cells, among 393 Q4 new-site events in those cells, report history selects 26.80 and misses 366.20. The proposed model selects 54.85 and misses 338.15. At equal shortlist size it gains 45.85 sites history missed, while dropping 17.80 that history selected: a net +28.05 report-site events [11].

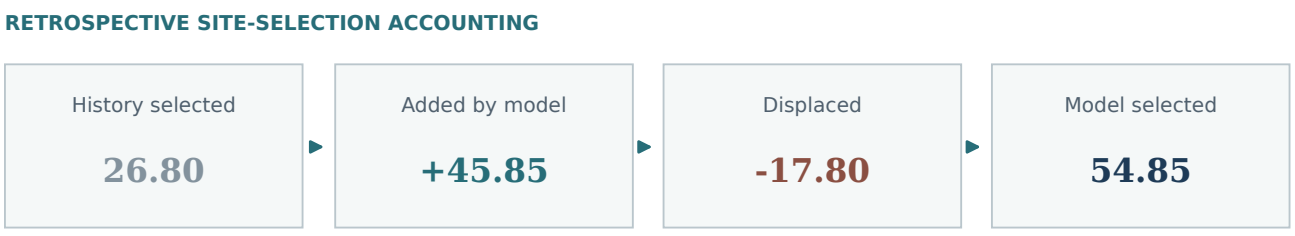


Figure 6. The model recovers some history-missed reported sites and gives up others. Counts split cutoff ties fractionally; they are not verified floods or prevented losses.

The economic question concerns the value of the added sites minus the value of displaced sites. It cannot assign either Table 1 loss scenario to all 366.20 sites missed by history: the model still misses most of them, and none has a measured avoidable loss.

7 Economic value at stake in overlooked areas

The top-5% unfamiliar-cell screen adds 45.85 later reported site-event equivalents and displaces 17.80 history picks. At THB 0.30m for every differently selected event, their illustrative exposure difference is THB 8.42m. Bangkok's dense commercial and transport areas can be worth far more per affected square. This arithmetic assumes each site-event is a distinct flooded square; the reports do not establish that.

A 2025 OpenStreetMap footprint overlay flags 2.85 added and 0.90 displaced site-event equivalents in cells with at least 20% mapped ground coverage [17]. Assign Table 1's THB 12m dense scenario to those flags and THB 0.30m elsewhere:

Added picks	Displaced picks	Difference
THB 47.10m	**THB 15.87m**	**THB 31.23m**

Figure 7. Conditional discovery-pool sensitivity, not measured damage or savings. Footprint coverage is a rough proxy: it omits storeys, tenants, road traffic, and unmapped buildings. Counts split shortlist ties fractionally.

7.1 When connected floods affect the wider city

Loss need not rise in proportion to flooded area. Several connected squares can close a through-route or delay access outside the wet footprint. Figure 8 starts with Table 1's THB 12m dense-square scenario and adds wider-network delay after a route is disrupted. It illustrates a threshold effect, not a measured Bangkok damage curve.

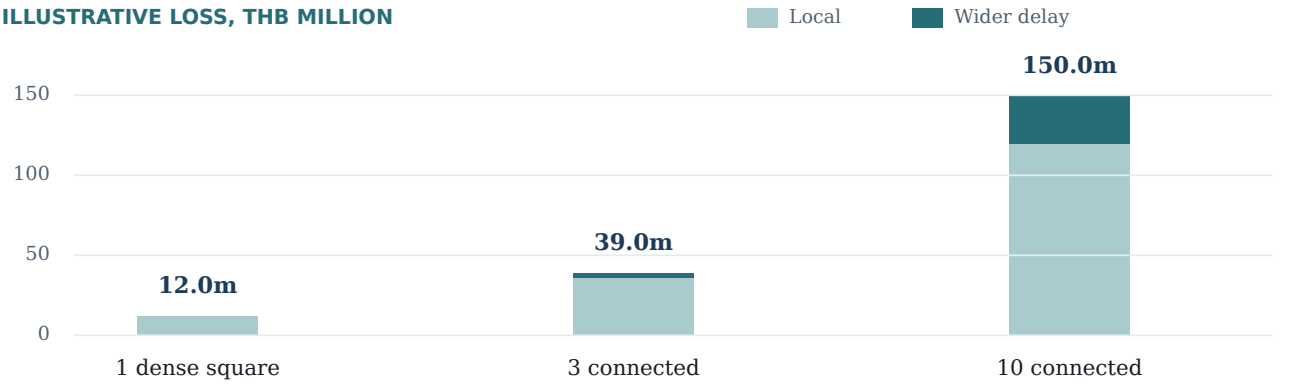


Figure 8. Worked dense-corridor scenario. Three squares add 20,000 person-hours of wider delay (THB 3m at THB 150/h); ten add 200,000 person-hours (THB 30m). These assumptions go beyond the local delays in Table 1.

The 45.85 added report sites are not known to form connected floods. Estimating real value requires verified extent, depth, duration, affected assets and travel, plus records of which early actions changed later outcomes and at what cost.

8 From triage to rainfall-conditioned planning

The present site-ranking CNN contains no rain channel. A follow-up experiment added five rainfall windows strictly before each decision time, using cached hourly district-point estimates [14]. That earlier fixed-allocation test showed mixed gains across one-hour and six-hour decisions; its capture values are not comparable with the quota-free curve in Section 5. The evidence does not establish a dependable rain-intensity response. Neither ranker can predict flood extent under twice the rain without a new outcome and validation design.

Table 2. Priority observations needed for a physical flood-extent and preparedness model.

Data class	Required observation
Storm forcing	Hourly gauge/radar rain, timing, forecast cutoff, antecedent wetness
Surface	Detailed terrain, barriers, land cover, road and inlet position
Drainage topology	Verified inlet-pipe-canal-outfall connections and capacity
Operating state	Dated pump/gate logs, discharge, canal/river/sea levels
Observed outcome	Timed urban wet and dry cells, depth and valid observation footprint

8.1 Mapped drainage assets are a starting point

A public BMA ArcGIS service supplies 81,574 geocoded storm-drain pipe polylines, as well as canal, pump, and gate layers [12]. The geometry does not verify every street inlet, pipe junction, outfall, or event-time pump/gate state. Some invert elevations suggest candidate gravity direction, but their datum and line ordering require validation. A static pipe map alone cannot explain whether a particular storm can drain on time.

8.2 A staged validation program

First, freeze the triage queue and measure later reported sites and field findings. Second, assemble strict time-matched rainfall, drainage state, and independent wet and dry observations; compare a rain-conditioned spatial model with simple land-cover/history rules and a hydraulic simulator on held-out storms and districts. Third, test scenario response to larger and longer storms with uncertainty ranges. Sentinel-1 radar can help with broad water extent under cloud, but narrow urban streets and valid dry observations need other sources [13].

9 Hackathon prototype and BMA collaboration

9.1 A map that makes the decision visible

The proposed hackathon demonstration lets a district choose a date and a fraction of unfamiliar cells to review, then compares two rankings: nearby-report history and the CNN. Reviewers can inspect each selected 250 m cell's report history, terrain, mapped roads, canals, and cover, and replay which later reported sites would have appeared within the same candidate fraction. The model, labels, and backtest are implemented; this interface is a proposed presentation layer, not a live flood alert.

9.2 What to test together

A useful BMA collaboration would begin in shadow mode in one or more districts. Before storms, freeze BMA's existing risk-point or inspection queue and the model queue. After field visits, log verified conditions, depth where measurable, likely cause, crew time, action taken, and time to resolution. The primary comparison is newly verified problem sites per fixed staff-hour budget; secondary measures include total verified sites, travel, and response time. BMA's current FY2569 risk-point targets provide the operational context, not proof of model impact [3].

9.3 Data partnership for the next capability

District feedback on false leads and overlooked sites would improve the triage model. Time-stamped radar or gauge rain, pump/gate operation, canal levels, and independently verified wet and dry locations would support a separate flood-extent model. That model would need held-out storms and districts, plus comparison with simple rules and a hydraulic simulator, before it could support larger-rain scenario planning.

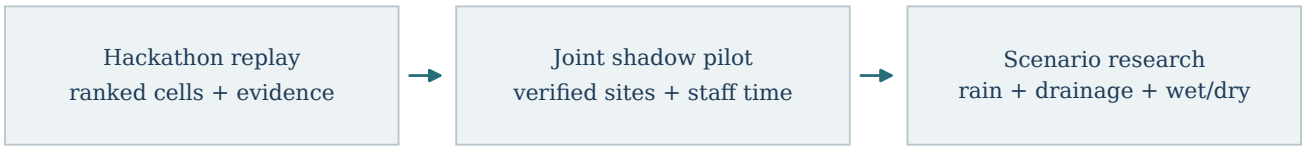


Figure 9. Proposed collaboration path. Each stage needs its own measured outcome before the next use is justified.

10 Conclusion and evidence boundary

The promising result is narrower than a flood forecast: among unfamiliar cells, spatial context raised mean rank of later reported new sites by 5.53 percentile points in held-out districts. A no-reserve full queue showed no gain at the top 5% of all cells, so the useful field allocation remains unresolved. Q4 and the 250 m definition were examined during model development; unreported cells are unknown, not proven dry. These limits make a prospective, field-verified BMA pilot the decisive next step. The hackathon offers a working ranking method and a concrete way to evaluate public benefit together.

References and reproducibility

[1] BMA, Traffy Fondue 2025 public report data. https://data.bangkok.go.th/dataset/traffy-fondue-2568 (accessed 24 Oct 2026). Grouped-site counts follow [8].

[2] BMA, FY2569 budget ordinance, DDS volume 69015, pp. 1 and 3. https://data.bangkok.go.th/dataset/2569 (accessed 24 Oct 2026).

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[4] FLOOD_CITYWIDE_HISTORY_RANKER.md and candidate_cells_by_district_250m.csv. Report-history benchmark and 10% shortlist-size audit.

[5] NASA Shuttle Radar Topography Mission (SRTM), elevation product; feature audit in FLOOD_CURRENT_STATE.md.

[6] OpenStreetMap contributors, road and waterway geometry; feature audit in FLOOD_CURRENT_STATE.md.

[7] ESA WorldCover 2021, land-cover product; feature audit in FLOOD_CURRENT_STATE.md.

[8] FLOOD_CITYWIDE_PHYSICAL_SITE_TEST.md and q4_2025_physical_site_final.json. Site labels, model arms, fold selection, intervals.

[9] FLOOD_CITYWIDE_SPATIAL_NOVEL_QUEUE.md and flood_citywide_spatial_novel_queue.py. Early CNN and nearby score.

[10] FLOOD_CITYWIDE_SPATIAL_MASK_AUGMENTATION.md. Earlier center and block-mask experiments.

[11] FLOOD_SITE_QUOTA_FREE_RANKING.md, q4_2025_site_novel_rank_curve.json, and q4_2025_site_novel_rank_curve_events.csv. Quota-free source selection, held-out ranking, site-level shortlist switches, and subgroup counts.

[12] FLOOD_BANGKOK_DRAINAGE_GIS.md. Audit of the BMA ArcGIS Thematic/Drainage service, including pipe and canal layers.

[13] FLOOD_CURRENT_STATE.md and FLOOD_OPEN_DATA_AUDIT_2026.md. Wider model and observation limits.

[14] FLOOD_CITYWIDE_SITE_RAIN_HOURLY.md. Matched pre-cutoff hourly-rain CNN experiment and five-fold results.

[15] FLOOD_SITE_ALL_CELL_CURVES.md and q4_2025_site_all_cell_curves.json. No-reserve full district-cell capture curves for all and emerging reported sites.

[16] FLOOD_SITE_DISTANCE_CURVES.md and q4_2025_site_distance_rank_curves.json. Emerging-site capture within unfamiliar cells by distance to earlier reports.

[17] osm_2025_building_footprint_citywide_250m.npz and accompanying audit JSON, derived by flood_citywide_surface_osm.py from the 1 Jan 2025 OpenStreetMap snapshot. Mapped ground-footprint sensitivity in Section 7.

Reproducibility. The generator build_flood_site_academic_paper.py reads the committed rank-curve JSON files, site-event CSV, and OSM footprint NPZ to rebuild this PDF. The model fitting and evaluation scripts are named in references [8,11]. Re-running those analyses requires source data listed in the repository README; the committed results alone reproduce the paper.

Repository: https://github.com/Plukio/flood-risk